



Noninvasive oral cancer screening based on local residual adaptation network using optical coherence tomography

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Abstract

Oral cancer is known as one of the relatively common malignancy types worldwide. Despite the easy access of the oral cavity to examination, the invasive biopsy is still essential for final diagnosis, which requires laborious operation and complicated trained specialists. With the development of deep learning, the artificial intelligence (AI) technique is applied for oral cancer examinations and alleviates the workload of manual screening on biopsy. However, existing computer-aided oral cancer diagnostic methods focus on oral cavity environment photos and histology images, which require complicated operations for doctors and are invasive and painful for patients. As a noninvasive, real-time imaging technique, optical coherence tomography (OCT) can express sufficient identical information for oral cancer screening, but it has not been effectively explored for automatic oral cancer diagnosis. This paper proposes a novel deep learning method named Local Residual Adaptation Network (LRAN) for noninvasive oral cancer screening on OCT images, collected from 25 patients in Beijing Stomatological Hospital. Our proposed LRAN consists of a Residual Feature Representation (RFR) module and a Local Distribution Adaptation (LDA) module. Specifically, RFR firstly adopts stacked residual blocks as the backbone network to learn feature representations for training data, optimized by the Cross-Entropy loss, and then deploy Euclidean distance to measure the distribution distance between training and testing OCT images. Finally, LRAN achieves distribution-gap bridging by the LDA module, which integrates local maximum mean discrepancy constraint to estimate and minimize the distribution discrepancy between training and testing sets within the same category. We also collected an OCT-based oral cancer image dataset to evaluate the effectiveness of the proposed method, and it achieves an accuracy of 91.62%, a sensitivity of 91.66%, and a specificity of 92.58% on this self-collected dataset. Furthermore, we conduct a quantitative and qualitative analysis, and the results demonstrate LRAN model has excellent capability to solve the noninvasive oral cancer screening task.

Keywords Noninvasive oral cancer screening · Deep learning · Automatic diagnosis · Residual local adaptation

1 Introduction

Oral cancer is one of the most widespread and deadly malignancies, and nearly one-third of patients come from underdeveloped regions and developing countries without

advanced diagnosing equipment and experts [1], leading severe fatality rate. Generally, the dentists first examine the patients' lip and mouth to look for abnormalities and then cut away the suspicious tissue samples for biopsy testing, which is still the primary diagnostic method for oral cancer. However, early oral cancer's clinical signs are not obvious, so it is laborious and time-consuming for doctors to perform pathological examinations, and biopsy testing is also painful and invasive to obtain screening samples from patients. It is necessary to develop an automatic diagnosis system to implement *noninvasive oral cancer screening*, with the flexibility to help experts reduce workload and save patients from invasive inspection.

In recent years, artificial intelligence (AI) has been increasingly applied in various automatic cancer diagnoses

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[2–4], as well as oral cancer [5–9] with the developments of deep learning technology. Nevertheless, most of existing automatic oral cancer detection researches focus on using mobile phone [5, 6] or histological samples [7–9] to capture images with deep learning. In contrast, the mobile-phone-based models achieve undesired accuracy and are susceptible to environmental disturbances in the oral cavity (e.g., teeth). Moreover, histological images are captured based on the invasive acquisition of oral tissues, making patients painful, and the tissue sampling procedure is time-consuming.

As we all know, optical coherence tomography (OCT) has excellent real-time, noninvasive, and label-free characteristics in body imaging. It has appeared to have effective examining performance in several practical applications [10–12], especially in bio-tissues analysis and disease diagnosis [13–15]. As a result, OCT has become a frequently-used oral tissue imaging technology and has been regarded as an effective indicator for oral cancer detection because it can generate high-resolution tissue microanatomy and cellular structure to observe the oral mucosa. More importantly, the OCT image generation is noninvasive and real-time, meaning that doctors do not have to perform a biopsy, and thus it can vastly reduce patients' pain using OCT screening.

Due to the advantages of OCT technology, several computer-aided methods exploit machine learning [16–19] on oral cancer diagnosis by quantitative analysis [17] and image processing [16, 18, 19] algorithm on OCT images. However, these models are sensitive to challenging scenarios, such as the variations of illumination, viewpoint, and imaging devices, which easily make the danger of misdiagnosis in another application.

To overcome the above drawbacks, this paper involves deep learning technology into noninvasive oral cancer screening, which has excellent capability to learn a robust identification model from large-scale OCT images to expand the usage scenarios for the proposed computer-aided system. Concretely, a novel convolutional neural network (CNN) architecture is proposed to conduct noninvasive oral cancer screening on self-collected 264 3-D OCT images (26,400 2-D images in total), and the network is named Local Residual Adaptation Network (LRAN).

The proposed LRAN model firstly releases each 3-D OCT image as 100 2-D images and then feeds them into a residual feature representation (RFR) module composing several stacked residual blocks to transform OCT images into CNN feature space. After that, a local distribution adaptation (LDA) module is integrated to alleviate the distribution discrepancy between training and testing sets within the same category, optimizing the identification model with a further adaptation on testing OCT images. Besides, we also collect 264 3-D OCT images to evaluate the effectiveness of our LRAN, and all data are from clinical patients of Beijing Stomatological Hospital. The experimental results and

extensive ablation study demonstrate the effectiveness and superiority of our proposed LRAN model on noninvasive oral cancer screening.

2 Related work

In recent years, significant progress has been made in computer-aided oral cancer diagnosis, including (invasive) deep learning-based [5–9], and (noninvasive) OCT-based [16–19] automatic cancer diagnosis.

2.1 Deep learning-based invasive oral cancer diagnosis

Successful deep learning applications inspired many researches in common cancer diagnosis [2–4] and several works integrated deep learning into oral cancer diagnosis, especially on mobile phone photos [5, 6] and histological tissue images [7, 9]. In detail, Welikala et al. [5] proposed a CNN method for automated oral cancer classification and lesion detection on mobile phone images using ResNet-101 and Faster R-CNN, and they further applied the transfer learning technique for binary oral cancer classification of “referral” vs. “non-referral” [6]. Das et al. [7] developed a two-stage oral cancer diagnosis approach for clinically relevant region identification using oral tissue histological images. Palaskar et al. [9] employed a transfer learning method to detect oral cancer using oral histological images under the microscope, which finally improved significantly with data augmentation techniques and fine-tuning compared with simple CNN baselines. In addition, popular oral cancer diagnosis methods are always developed from four CNN architectures, e.g. VGG [20], GoogLeNet (Inception-V1) [21], Inception-V3 [22], and ResNet [23]. Concretely, the VGG [20] network is a classical convolutional neural network architecture with multiple layers, which is the basis of ground-breaking object recognition models. GoogLeNet (Inception-V1) [21] is developed from “Network in Network” and inception module by adding 1×1 convolution at the middle of the network and having different sizes/types of convolutions for the same input and stacking all the outputs, which is quite different from VGG network. Inception-V3 [22] is a CNN architecture from the inception family that makes several improvements, including using label smoothing, factorized 7×7 convolutions, and auxiliary classifier to propagate label information lower down the network. ResNet [23] utilized skip connections or shortcuts to jump over some layers with double- or triple-layer skips that contain nonlinearities and batch normalization, performing excellent results in different computer vision tasks [24–27].

Though the above-mentioned deep learning methods performed impressive effectiveness on automatic oral cancer

detection, there also exist two challenging limitations in these models. The first one is that the methods based on mobile phone photos [5, 6] are severely restricted by oral environments, such as illumination, occlusion, and viewpoints, so that these models will generate poor performance when switching to another scenario. The second limitation is the tissue sampling procedure of histological images, whose performance is painful and invasive for patients and complicated and time-consuming for professional doctors. Therefore, developing an automatic noninvasive oral cancer diagnosis system is vital.

2.2 OCT-based noninvasive oral cancer diagnosis

As a popular noninvasive imaging technology for disease screening, optical coherence tomography (OCT) is a perfect object to achieve noninvasive oral cancer diagnosis. For OCT-based oral cancer detection, several early researches [16–19] have already implemented quantitative analysis and image processing algorithms on OCT oral images. Specifically, Lee et al. [16] had studied the normal oral mucosa of 54 volunteers, 39 patients with mild dysplasia, and 44 patients with moderate dysplasia using SS-OCT with computer-aided analysis of the final images, and recognized all lesions with a sensitivity of 82% and specificity of 90%. Yang et al. [17] identified oral cancer by investigating the quantitative difference of cancer and non-cancer tissues, and an optical attenuation model and oral tissue OCT images from 14 patients were used in their study, achieving a sensitivity of 97.88% and a specificity of 83.77%. Heidari et al. [18] designed a mobile OCT imaging system using an image

processing algorithm, and the point-of-care oral diagnosis performed at or exceeded the performance of visual observer scoring of OCT images. Sunny et al. [19] demonstrated that intraoperative OCT with an automated diagnostic algorithm could improve the accuracy of surgical margin delineation compared to the existing methods.

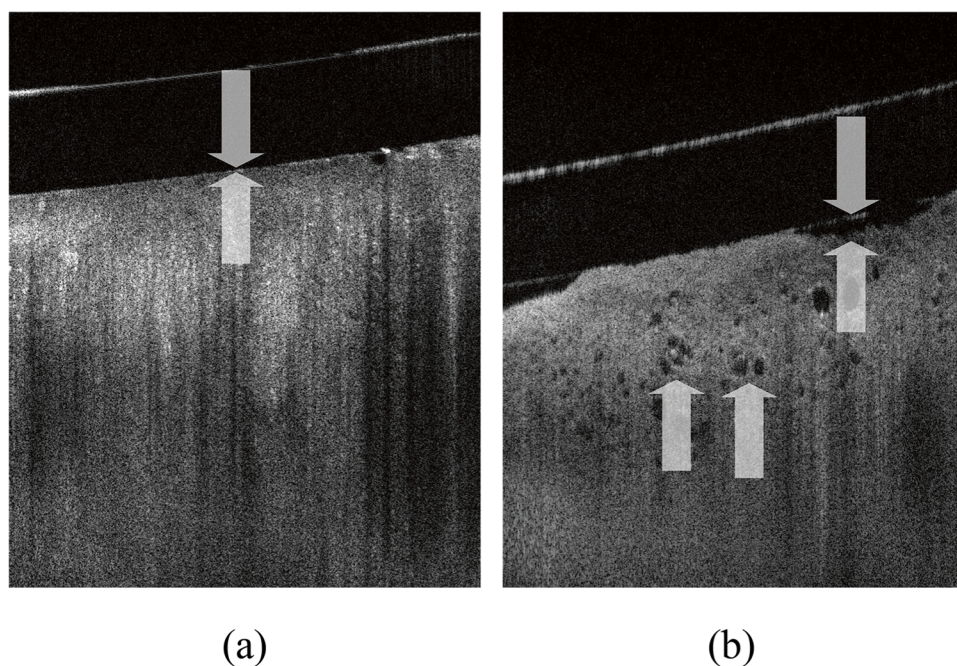
However, those quantitative image analysis and traditional image processing methods only work well on small data in specific scenarios (e.g. normal illumination, high resolution, or clear contrast). But they always perform with poor results under changeable practical environments (e.g. poor illumination, low resolution, or blurry contrast). Thus, we aim to develop an accurate, low-cost, and efficient deep learning-based noninvasive cancer screening approach on OCT images to solve the challenges discussed above.

3 Materials and methods

3.1 Data acquisition

This paper collects an oral OCT image dataset from Sep 2020 to Feb 2021 in Beijing Stomatological Hospital, Capital Medical University, China. It contains 264 3-D OCT images from 25 patients, and each 3-D image can be released into 100 2-D images. Ethics committee approval and institutional review board exemption are obtained, and patients signed the informed consent. In detail, there are 120 non-cancerous OCT images and 144 cancerous (Fig. 1). As for OCT imaging, we deploy an ultra-high-resolution microscopy imaging system, with 300-nm spectral bandwidth,

Fig. 1 OCT images obtained from the oropharynx (a) and tongue (b): **a** The epithelial layer, basement membrane, and lamina propria are clearly identified, and the tissue layers are well-arranged. **b** There are no normal squamous epithelial structure and basement membrane, the overall refraction is uneven, and nestlike dark areas are clear



800-nm center wavelength, and 1.5- μm axial resolutions in tissue supercontinuum fiber laser source, to capture data from patients. Besides, the light source spectrum is split by a 50:50 coupler and introduced into the reference and sample arms. Using $\times 4$ objective lens in the sample arm, the light penetration depth and the lateral resolution can reach about 2 mm and 3.5 μm , respectively. The imaging system scanned multiple patches of oral tissues immediately after surgery resection, including the tongue, floor of the mouth, upper and lower gingiva, buccal mucosa, cheek pouch, and oropharynx, ranging from early to terminal stages.

In experiments, two senior dental pathologists accomplished the annotation work of OCT images, and the diagnosing standards are summarized in Table 1. Moreover, all 3-D OCT images are released into 26,400 2-D images, which are divided into training and testing sets according to Table 2.

3.2 Method

Inspired by recent researches on computer-aided diagnosis, we utilize Convolutional Neural Network (CNN) to achieve the noninvasive oral cancer diagnosis task because CNN has excellent capability to extract identical feature representations for large scale data automatically. In detail, we adopt a residual Network to learn residual CNN appearance features from OCT images and then improve the scalability to testing data by a proposed local distribution adaptation method. The overall illustration can be found in Fig. 2.

3.2.1 Residual feature representation module

As a representative and effective mean of deep learning, the convolutional neural network can transform input images into feature vectors by successive convolutional layers and pooling layers with iterative filters and then predict the category according to the distinguishable information from feature vectors. In detail, CNN can map each image into hierarchical feature maps, learning discriminative features at varying spatial levels without the need for manually tuned parameters.

Table 2 The distribution of training and testing data

Dataset	Non-cancerous	Cancerous	Total
Training	6000	7900	13,900
Testing	6000	6500	12,500
Total	12,000	14,400	26,400

In this paper, we introduce residual block [23] into our model, one of the most powerful CNNs, and its representative framework ResNet has achieved excellent performance in various image classification tasks [24–27]. Specifically, the residual block is a stack of layers by adding the output of a layer to another deeper in the block (Fig. 3), which is mathematically defined by,

$$\mathbf{x}_l = \text{conv}_l(\mathbf{x}_{l-1}) + \mathbf{x}_{l-1} \quad (1)$$

where conv_l represents the l -th convolutional layer in the networks, $\mathbf{x}_l$ denotes the output of the l -th layer. The Residual block allows the information to flow from the initial layers to the last layer. It can protect the deep networks with many parameters from vanishing gradient problem, so that learning more discriminative features for oral OCT images.

To exploit the noninvasive oral cancer classification task, this paper employs a ResNet-50 framework involving 16 residual blocks, average pooling, and fully connected layers to implement the residual feature representation using OCT images. The detailed architecture can be found in [23], and the kernel sizes involved in 16 residual blocks are summarized in Table 3.

3.2.2 Local distribution adaptation module

For the noninvasive oral cancer screening task, the N OCT images are divided into N_{tr} training and N_{te} testing images. Specifically, the training data is defined by $\mathcal{D}_{tr} = \{\mathbf{x}_i^{tr}, y_i^{tr}\}_{i=1}^{N_{tr}}$, where $\mathbf{x}_i^{tr}$ represents the i -th image in the training set, and y_i^{tr} is the annotation for $\mathbf{x}_i^{tr}$; The testing data is a sequence of unlabeled oral OCT images $\mathcal{D}_{te} = \{\mathbf{x}_j^{te}\}_{j=1}^{N_{te}}$. Our oral cancer

Table 1 Diagnosing standards for oral OCT images

Diagnosis	Standard
Non-cancerous	(1) Uniform refraction (2) The squamous epithelium arranged in order with a reticular structure (3) Obviously stratified between epithelial cell layer and lamina propria with the intact basement membrane
Cancerous	(1) Invisible basement membrane (2) No stratified structure of epithelium and stroma (3) Disordered structure (4) Irregular refraction or cancer nest

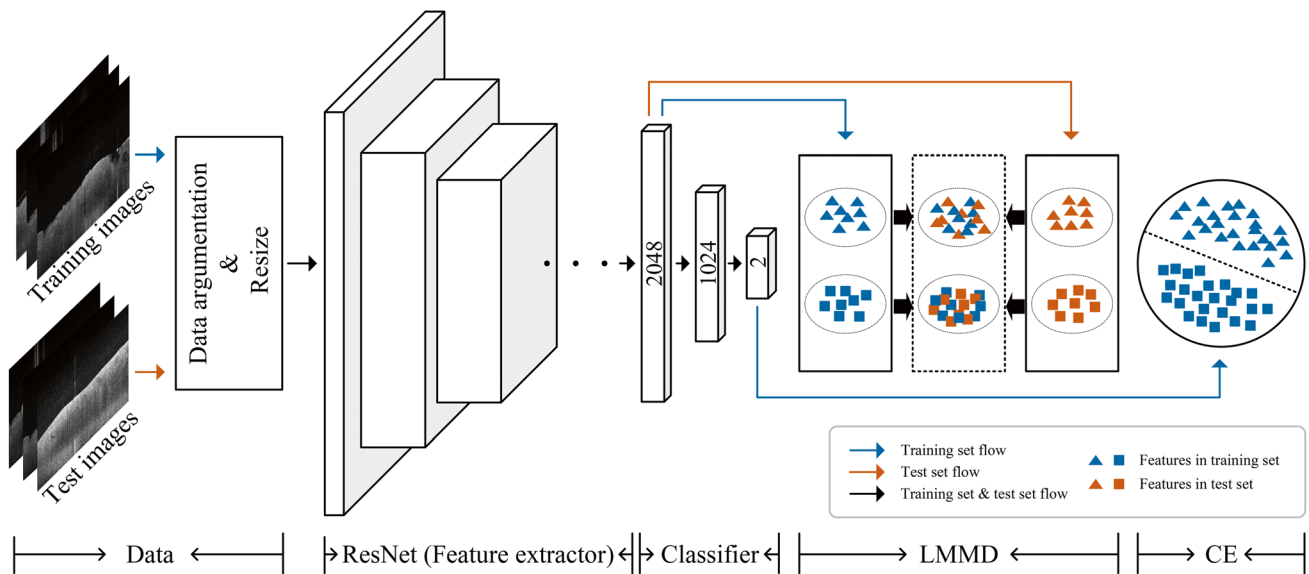
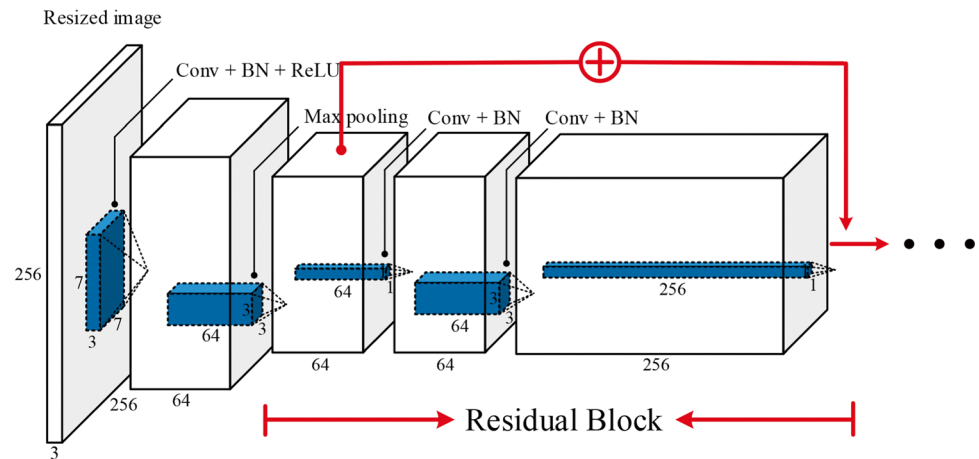


Fig. 2 The overview of LRAN. The image features are extracted by the proposed residual feature representation module after data argumentation and resizing. Then, the distribution of training set and test set in the same category is drawn close by minimizing the $\mathcal{L}_{lmm}$.

Finally, we utilize the output predicted probabilities of the classifier to align LRAN with the distribution of the training set by cross-entropy (CE) loss

Fig. 3 The architecture of residual block. The red line denotes skip connection, which can reduce the impact of the vanishing gradients problem



screening task aims to predict the labels $\{y_j^{te}\}_{j=1}^{N_{te}}$ for testing set, where $y_i^{tr}, y_j^{te} \in \{0, 1\}$, and “0” denotes normal (non-cancerous), and “1” denotes abnormal (cancerous). In detail, the ResNet-50 in our LRAN is decomposed by a feature extractor $\mathcal{F}(\cdot)$ and a classifier $P(\cdot)$. Given an OCT image $\mathbf{x}$, the feature extractor firstly learns its discriminative feature vector $\mathcal{F}(\mathbf{x}) \in \mathbb{R}^d$, where $d = 1024$ is the dimension of the feature vector, and then the feature vector is fed into the classifier to obtain its prediction $P(\mathcal{F}(\mathbf{x})) \in \mathbb{R}^C$, where C denotes the number of classes in OCT images.

Firstly, we conduct the network optimization from training data $\mathcal{D}_{tr} = \{\mathbf{x}_i^{tr}, y_i^{tr}\}_{i=1}^{N_{tr}}$ by a classification task,

achieved by cross-entropy (CE) loss. CE is often utilized to measure the global difference between two probability distributions, and it is developed from information theory. In our noninvasive oral cancer classification task, the identification ability of our network is essentially revealed from the dominated performance on OCT images. Thus, the cross-entropy loss reduces the differences between the model’s predicted space and the real label space of training data, improving the classification performance.

Therefore, the cross-entropy loss in this paper is mathematically calculated by,

Table 3 The kernel sizes of 16 residual blocks in our convolution operation

Layer name	Output size	Kernel size
Conv1	128×128	7×7 , stride 2 3×3 max pool, stride 2
Conv2_x	64×64	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 364 \\ 1 \times 1256 \end{bmatrix} \times 3$
Conv3_x	32×32	$\begin{bmatrix} 1 \times 1128 \\ 3 \times 3128 \\ 1 \times 1512 \end{bmatrix} \times 4$
Conv4_x	16×16	$\begin{bmatrix} 1 \times 1256 \\ 3 \times 3256 \\ 1 \times 11024 \end{bmatrix} \times 6$
Conv5_x	8×8	$\begin{bmatrix} 1 \times 1512 \\ 3 \times 3512 \\ 1 \times 12048 \end{bmatrix} \times 3$

$$\mathcal{L}_{ce} = -\mathbb{E}_{\{\mathbf{x}_i^{tr}, y_i^{tr}\} \in \mathcal{D}_{tr}} \sum_{c=1}^C q_c^{tr} \log P_c(\mathcal{F}(\mathbf{x}_{tr})) \quad (2)$$

where $C = 2$ is the number of classes, q_c^{tr} denotes the c -th element in the one-hot encoding q^{tr} of y^{tr} , where $q^{tr} = [1, 0]$ means $y^{tr} = 0$, $q^{tr} = [0, 1]$ means $y^{tr} = 1$, $P_c(\cdot)$ means the c -th element output of the classifier (the predicted probability for c -th class).

From the model optimization among labeled training images, the prediction model has sufficiently learned the distribution probability of training data. However, it will lose accuracy when it is applied to the unlabeled testing images due to the distribution discrepancy between training and testing sets caused by various imaging influences (e.g. illumination, viewpoint, or tissue position). Namely, cross-entropy loss only focuses on aligning the model distribution to the global training set without considering the relationships between training and testing sets to make irrelevant data too close to be classified accurately.

Therefore, this paper develops a local distribution adaptation module to further train the network by alleviating the distribution discrepancy between training and testing sets. As a primary method to bridge the distribution gap, maximum mean discrepancy (MMD) [28] is often utilized for pulling the global average features from two sets as much as close, and the distribution discrepancy reduction between training and testing sets can be achieved by minimizing MMD cost,

$$\mathcal{L}_{mmd} = \left\| \frac{1}{N_{tr}} \sum_{\mathbf{x}_i^{tr} \in \mathcal{D}_{tr}} \mathcal{F}(\mathbf{x}_i^{tr}) - \frac{1}{N_{te}} \sum_{\mathbf{x}_j^{te} \in \mathcal{D}_{te}} \mathcal{F}(\mathbf{x}_j^{te}) \right\|^2 \quad (3)$$

The MMD cost can keep the global relationship among training and testing sets close. Nevertheless, it does not pay attention to the local category correlations between training and testing images within the same class, such that it has a limited positive impact on avoiding the irrelevant data too close to be predicted accurately. To overcome this drawback in MMD cost, we develop an improved local distribution adaptation module based on Local Maximum Mean Discrepancy (LMMD) [29], which can extend the ability of model adaptation by capturing the fine-grained information for each category. Thus, we utilize LMMD to measure the cost of the local discrepancy between training and testing feature spaces and minimizing the local distribution adaptation loss by,

$$\mathcal{L}_{lmmd} = \frac{1}{C} \sum_{c=1}^C \left\| \sum_{\mathbf{x}_i^{tr} \in \mathcal{D}_{tr}} w_i^{tr,c} \mathcal{F}(\mathbf{x}_i^{tr}) - \sum_{\mathbf{x}_j^{te} \in \mathcal{D}_{te}} w_j^{te,c} \mathcal{F}(\mathbf{x}_j^{te}) \right\|^2 \quad (4)$$

where $w_i^{tr,c}$ and $w_j^{te,c}$ denote the weights of $\mathbf{x}_i^{tr}$ and $\mathbf{x}_j^{te}$ belonging to class c , respectively. Note that, $\sum_{i=1}^{N_{tr}} w_i^{tr,c} = 1$ and $\sum_{j=1}^{N_{te}} w_j^{te,c} = 1$, $\sum_{\mathbf{x}_i \in \mathcal{D}} w_i^c \mathcal{F}(\mathbf{x}_i)$ is a weighted sum on class c .

In Eq. 4, the weight $w_i^{tr,c}$ of the training sample $\mathbf{x}_i^{tr}$ in OCT images can be calculated by,

$$w_i^{tr,c} = \frac{q_{i,c}}{\sum_{(\mathbf{x}_k^{tr}, y_k^{tr}) \in \mathcal{D}_{tr}} q_{k,c}^{tr}} \quad (5)$$

where $q_{i,c}^{tr}$ is the c -th one-hot encoding element of y_i .

In contrast, the weight $w_j^{te,c}$ of the testing image $\mathbf{x}_j^{te}$ is mathematically formulated by,

$$w_j^{te,c} = \frac{q_{j,c}^{te}}{\sum_{(\mathbf{x}_k^{te}) \in \mathcal{D}_{te}} q_{k,c}^{te}} \quad (6)$$

where $q_{j,c}^{te}$ is the c -th element of the predicted probability vectors $P(\mathcal{F}(\mathbf{x}_j^{te}))$.

As the calculation in Eq. 4, the optimization of $\mathcal{L}_{lmmd}$ can minimize the local distribution discrepancy between the training set and test set, so that pulls the similar samples possibly belong to the same category into a close feature space, as illustrated in Fig. 2.

Overall, the final loss function of our proposed LRAN model is formulated by,

$$\mathcal{L} = \mathcal{L}_{ce} + \alpha \mathcal{L}_{lmmd} \quad (7)$$

where α is the balanced parameter of the cross-entropy loss and the LMMD loss.

4 Experiments

This section conducts comprehensive experiments to evaluate our Local Residual Adaptation Network (LRAN) on the noninvasive oral cancer classification task. The detail of experimental settings and evaluation metrics is firstly described, and then the results and ablation studies are analyzed to prove the effectiveness of our LRAN model.

4.1 Experimental settings

In our LRAN network, ResNet-50 is adopted as the backbone network for feature extraction, initialized by pre-train parameters from ImageNet [30]. All images are resized to 256×256×3, and data augmentation is performed by applying random horizontal flip. In the training stage, we utilize Adam [31] optimizer to update network parameters, with a learning rate 1×10^{-5} , multiplied 0.1 within every 20 epochs. Each batch selects 32 images from training and testing sets, and the balance parameter in Eq. 7 is gradually changed from 0 to 1 by a progressive schedule $\alpha = 2/\exp(-10\theta) - 1$ (θ is the training progress linearly changing from 0 to 1) to suppress noisy activations at the early training stage. Besides, our experiments are implemented by the PyTorch framework on 2×NVIDIA RTX 2080Ti GPUs. As for the training time, each epoch costs 73.07 s with a total of 40 epochs, and the time cost per training image is 5.3 ms.

4.2 Evaluation metrics

To evaluate our performance, we adopt accuracy, sensitivity (or true positive rate, TPR), specificity, and the area under the receiver operating characteristic (AUROC, or AUC) as metrics. In detail, accuracy is the overall percentage of images that are identified correctly; sensitivity is the proportion of abnormal images classified correctly, while specificity is the proportion of normal images classified correctly. Both sensitivity and specificity represent the identification capability for oral cancer in OCT images. More concretely,

the evaluation metrics are mathematically formulated as below:

$$\begin{aligned} \text{Accuracy} &= \frac{TP+TN}{TP+TN+FP+FN} \\ \text{Sensitivity} &= \frac{TP}{TP+FN} \\ \text{Specificity} &= \frac{TN}{TN+FP} \end{aligned} \quad (8)$$

where TP means true positive, representing the number of positive (cancerous) images that are correctly identified; TN means true negative to represent the number of negative (non-cancerous) images that are correctly classified; FP is *false positive*, denoting the number of negative images that incorrectly indicate the presence of positive, and FN is false negative, indicating the number of positive images incorrectly identified negative.

4.3 Ablation studies

4.3.1 Compared with popular CNN architectures

In order to compare the effectiveness of our LRAN method with other deep learning architectures, we employ four popular CNN frameworks to conduct oral cancer identification on OCT images, including VGG-16 [20], GoogLeNet(Inception-V1) [21], Inception-V3 [22], and ResNet-50 [23]. The comparative results are summarized in Table 4.

As observed in Table 4, our LRAN achieves an accuracy of 91.62%, sensitivity of 91.66%, and specificity of 92.58%. Compared with popular networks, the highest results are the accuracy of 91.62% (our LRAN), a sensitivity of 93.42% (ResNet-50), and a specificity of 92.58% (our LRAN). In a word, our LRAN method achieves the best performance on accuracy and specificity metrics and obtains the second-highest result on sensitivity. That demonstrates LRAN approach has an overall excellent identification capability for oral cancer over optical coherence tomography images.

Table 4 Results of LRAN and compared methods

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)
VGG-16	88.41	91.08	85.52
GoogLeNet	86.54	88.34	84.60
Inception-V3	87.90	91.34	84.17
ResNet-50	88.65	93.42	83.48
GRAN(Ours+MMD)	89.77	90.80	88.65
LRAN	91.62	91.66	92.58

The overall accuracy, sensitivity, and specificity are calculated. The best results are highlighted in boldface, and the second-best results are italicized

4.3.2 Evaluation with global adaptation

To validate the adaptation ability of our proposed local adaptation module, we integrate global adaptation into our network, named Global Residual Adaptation Network (GRAN), which involves Maximum Mean Discrepancy [28] to achieve adaptation, instead of LMMD.

Specifically, MMD [28] is a measurement for distribution gap over average feature vectors from different sets, without considering category division in each set, which is defined by,

$$\left\| \mathbb{E}_{\mathbf{x}_i^{tr} \in \mathcal{D}_{tr}} [\mathcal{F}(\mathbf{x}_i^{tr})] - \mathbb{E}_{\mathbf{x}_i^{te} \in \mathcal{D}_{te}} [\mathcal{F}(\mathbf{x}_i^{te})] \right\|^2 \quad (9)$$

As the formula implies, MMD focuses on the alignment of the global distributions, ignoring the relationships between two subsets within the same category. To verify the importance of the relevant subsets within the same category between training and testing sets, we compare the performances of LRAN with LMMD and GRAN with MMD.

From Table 4, we can observe that GRAN achieves 89.77% accuracy, 90.80% sensitivity, and 88.65% specificity, which has a non-neglectable distance (1.85% accuracy, 0.86% sensitivity, and 3.93% specificity) to LRAN. Compared with GRAN, the improvements overall evaluation metrics prove the effectiveness of exploiting the local correlation between training and testing sets by our LRAN.

4.3.3 Evaluation of network with different depths

The previous discussion reveals the potential capability of deep learning in OCT oral cancer screening, and here we explore the influence of the network depths for OCT feature representation. In detail, ResNet is deployed as the feature extractor in our LRAN network, and we implement the network by using different architectures of ResNet, including 18, 34, and 50 layers. As shown in Table 5, the accuracy results of LRAN with 18, 34, and 50 layers are 89.62%, 90.90%, and 91.62%, respectively. It is obvious that the performance is improved with the increasing of network layers. With the layers increasing, we can conclude that the network can exploit more discriminative information from OCT images to identify oral cancer.

Table 5 Results of LRAN with different network depths

Depth	Accuracy (%)	Sensitivity (%)	Specificity (%)
18	89.62	88.51	89.73
34	<i>90.90</i>	<i>90.37</i>	<i>91.48</i>
50	91.62	91.66	91.58

The best results are highlighted in boldface, and the second best results are italicized

4.4 Visualization analysis

4.4.1 Confusion matrix

The confusion matrix is a table that is often used to describe the performance of a classification model. In a confusion matrix, each row represents the real class, while each column denotes the predicted class. In order to reveal the superiority of our LRAN model, we visualize the confusion matrices for compared methods and our LRAN model in Fig. 4. As illustrated in Fig. 4, 11,202 out of 12,500 images are correctly identified by LRAN-18, 11,363 out of 12,500 images are correctly identified by LRAN-34, and LRAN-50 correctly identifies 11,453 out of 12,500 images. These results show that deep learning methods, especially LRAN, can automatically detect oral cancer in OCT images with a low error, and our LRAN achieves the best performance over compared methods.

4.4.2 Receiver operating characteristic curve

The Receiver Operating Characteristic (ROC) curve is a graph showing the performance of a classification model and is commonly used for binary classification problems. It is created by plotting the sensitivity against the (1 - specificity) at all classification thresholds. The Area Under the Receiver Operating Characteristic (AUROC, or AUC) is a performance measurement based on the ROC curve, which is also an import evaluation metric for a classification model. Generally, the more “up and to the left” the ROC curve is, the higher the AUC and the better the classification model. The ROC curve of LRAN is generated, and AUCs are calculated in Fig. 5. As observed in Fig. 5, it is close to the upper-right corner with an AUC of 0.902, which demonstrates our LRAN achieves a performable classification model on non-invasive oral cancer screening task using OCT images.

4.4.3 T-distributed stochastic neighbor embedding visualization

To represent the feature learning ability of our network, we adopt T-Distributed Stochastic Neighbor Embedding (t-SNE) to visualize the feature representations of testing images. The t-SNE is an unsupervised, non-linear technique primarily used for data exploration and visualizing high-dimensional data. In simple terms, t-SNE brings us a feel or intuition of how the data is arranged in a high-dimensional space. For an excellent classification model, extracted features in different categories can be well separated after dimension reduction using t-SNE. To evaluate the performance of compared methods and our LRAN intuitively, we generate t-SNE plots in Fig. 6 for each model. As shown in Fig. 6, the features extracted

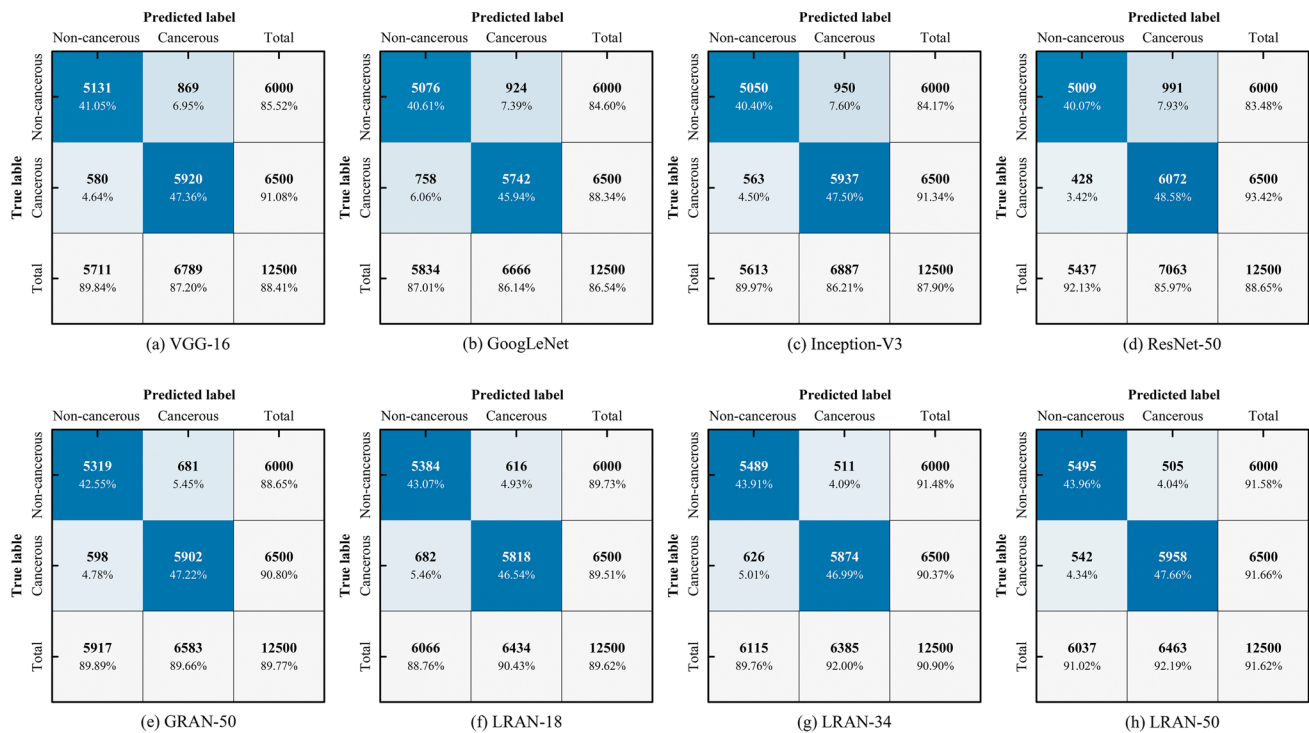


Fig. 4 The confusion matrices of compared methods and our LRAN

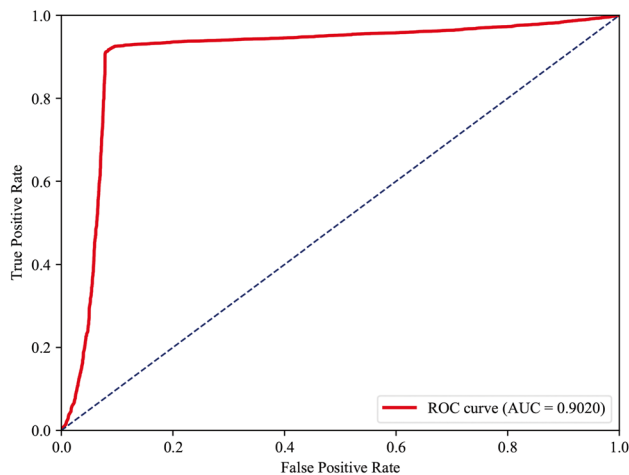


Fig. 5 ROC curve of LRAN and the AUC is calculated

from LRAN-50 are visualized in the best separation, while others are less overlapping among different classes. In this case, it is reasonable to summarize that our LRAN can explore discriminative feature representations from non-invasive OCT images for the oral cancer screening task.

4.4.4 Loss and accuracy variations

The goal of deep learning is to obtain a convergent model that fits the statistical distribution of testing data. Essentially, by minimizing the objective function (loss function) using the optimization algorithm, the model will approach the distribution step by step. Typically, as the loss decreases, the classification model is closer to the distribution and achieving higher accuracy. As a result, we visualize the loss and accuracy variations for each compared method and our LRAN along with training steps in Fig. 7. As demonstrated in Fig. 7, all the losses are rapidly decreased at a low level, and the accuracies also achieve a high performance in a short time. In detail, the compared methods achieve a lower loss (1×10^{-5}), but they do not reach a higher accuracy at the end of the training process. Besides, the losses of LRANs are much lower than GRAN (1×10^{-3} vs. 2×10^{-2}), but their accuracies are higher, which indicates that LMMD can better estimate the distribution discrepancy between training and testing sets. Thus, it can effectively improve the screening performance for oral cancer in OCT images.

4.4.5 Analysis for misclassified data

To analyze the classifying capability of our LRAN model, we group the wrongly classified data and visualize representative samples in Fig. 8. It can be observed that the misclassified samples from either non-cancerous or cancerous contain noticeable noises, and these images do not

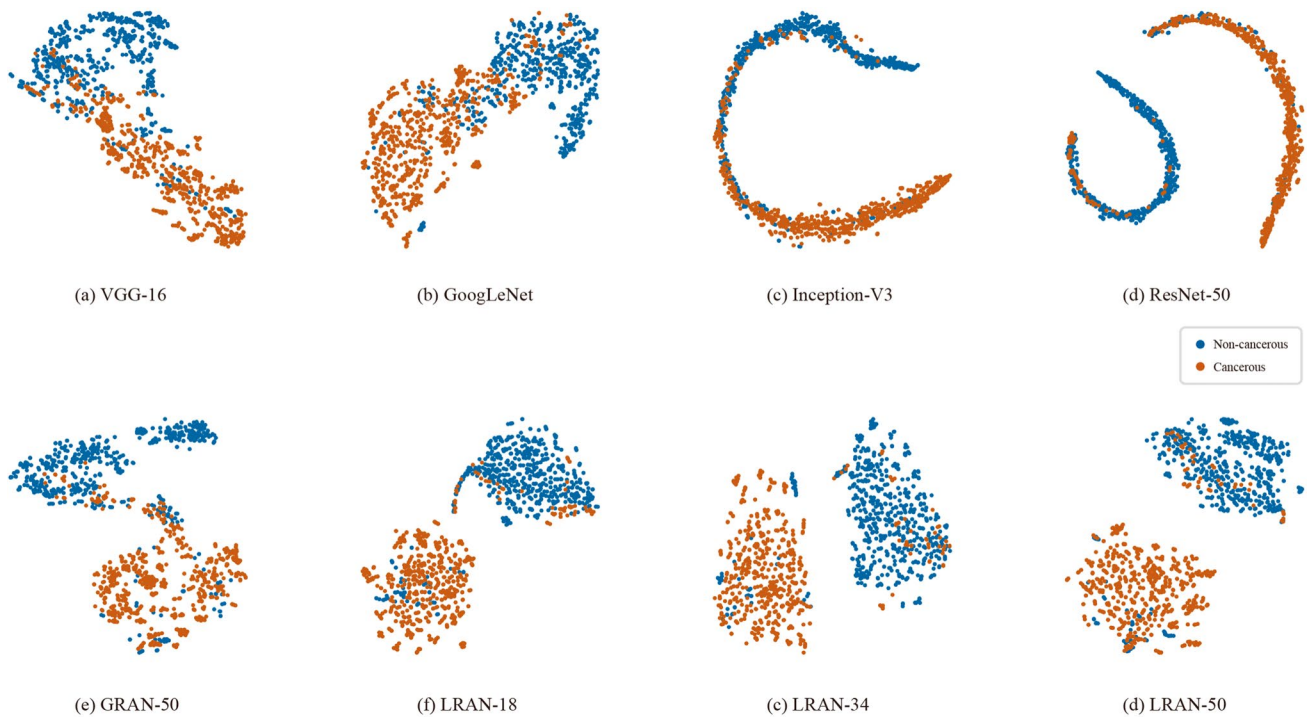


Fig. 6 T-SNE plots of compared methods and our LRAN

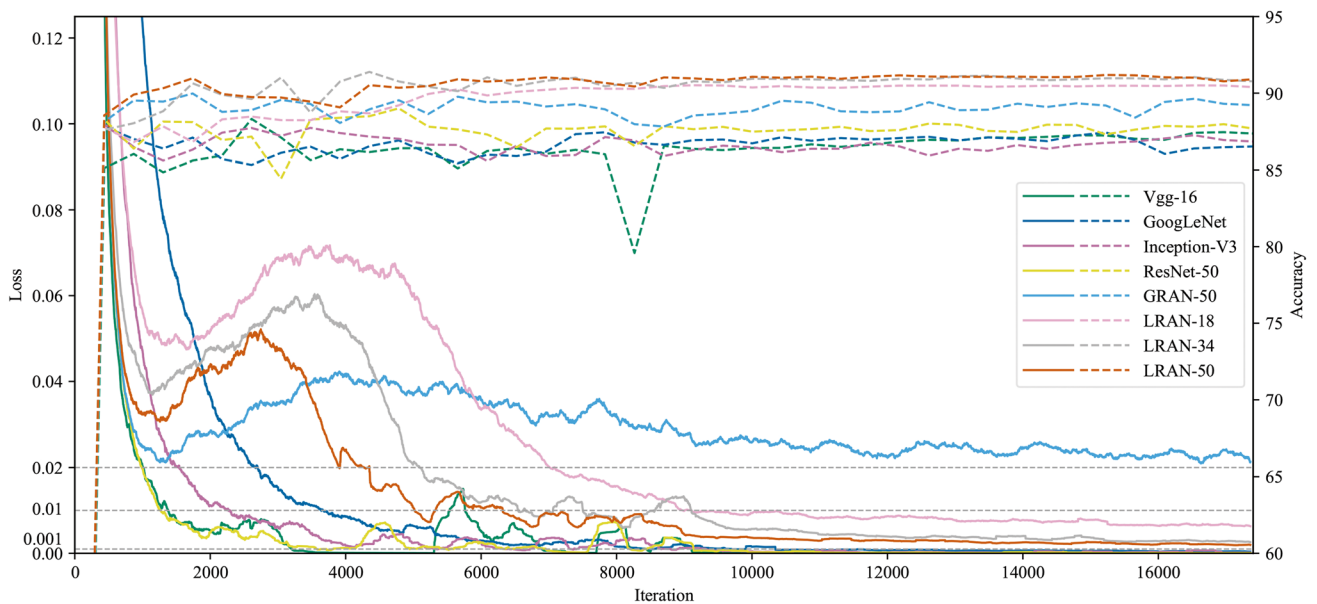
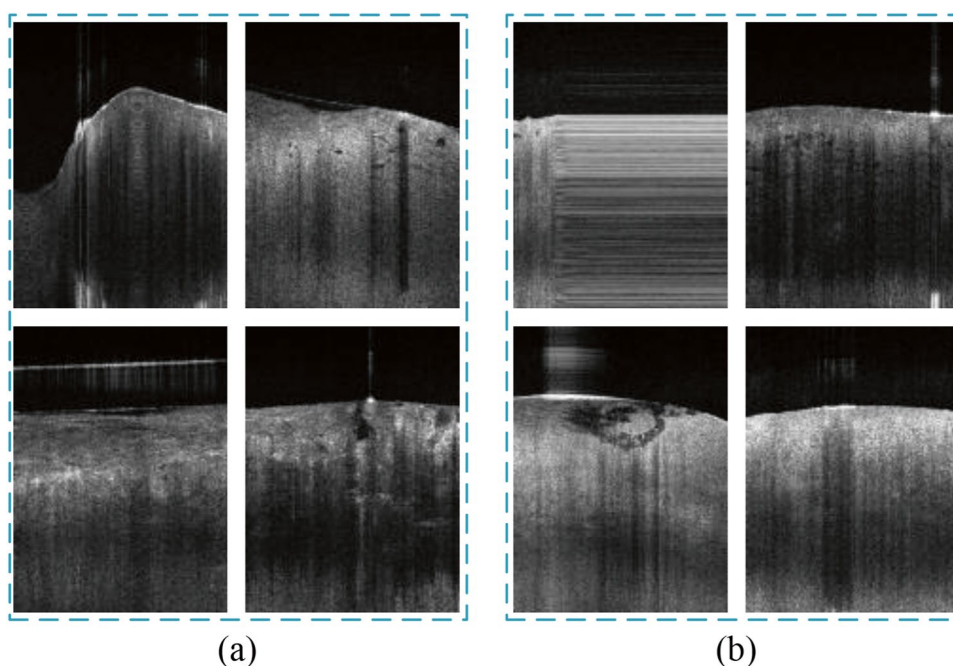


Fig. 7 Loss and accuracy variations of our LRAN method. The solid line represents the loss, and the dashed line represents the accuracy. The losses are calculated using the average loss of the last epoch (435 iterations), and the accuracies are calculated every epoch (40 times)

have obvious biological signs for oral cancer, which may be even confusing to the human experts in clinical diagnosis. Therefore, these challenging OCT images should be discarded in data collection due to the noticeable noises. In our future work, we will introduce an image quality

assessment algorithm to measure the OCT images for collecting high-quality data to address the above challenges, training a more robust model for the AI-based noninvasive oral cancer diagnosis task.

Fig. 8 Samples of wrongly-classified data. **a** The misclassified non-cancerous images. **b** The misclassified cancerous images



5 Conclusion and future works

Aiming to solve noninvasive oral cancer screening task using deep learning technology, this paper proposes a novel Local Residual Adaptation Network (LRAN) on optical coherence tomography images. The LRAN network is achieved by a residual feature representation module and a local distribution adaptation module, which can alleviate the distribution discrepancy between training and testing data in the CNN feature level. Furthermore, the quantitative and qualitative analyzing experiments evaluate our proposed LRAN method by comparing it with popular networks and modified LRAN, and the results prove the effectiveness on the noninvasive oral cancer screening task. With the help of deep learning, we can develop an accurate and low-cost automatic oral cancer identification system in a noninvasive manner without depending on clinicians to examine OCT images manually. As for the significant limitations in practice, the small-scale data and the requirement of high-quality images often constrain the performance of deep learning methods because sufficient model training needs large-scale, high-quality images. Compared to existing methods, our proposed LRAN can exploit more discriminative information with better results than them because the local distribution adaptation loss can mine the inherent correlations between training and testing sets.

In future, we will develop our LRAN model to solve the oral cancer staging problem on noninvasive OCT images. Different from the binary diagnosing task, it stages each OCT image into multiple categories, containing more discriminative clues than binary diagnosis. As for our LRAN, it has excellent capability to exploit local

discriminative information from each class by utilizing CNN and local maximum mean discrepancy model to bridge distribution gap. Therefore, our LRAN can be successfully transferred into the oral cancer staging task with satisfying performance, and we will introduce an attention mechanism to enhance LRAN for extracting much more discriminative features from OCT images.

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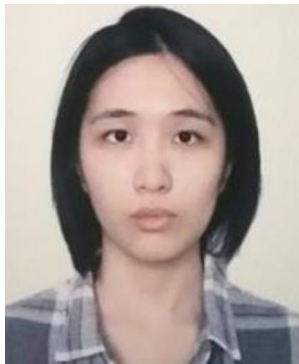
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